

Gourav Prateek SHARMA

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RESEARCH INTERESTS

Time-sensitive Networking (TSN) and Deterministic Networking (DetNet) and their integration with wireless technologies (e.g., WiFi and 5G URLLC).

EXPERIENCE

National Institute of Technology Kurukshetra *July 2025 - Present*
Assistant Professor, Dept. of ECE

Thapar Institute of Engineering and Technology *Dec 2024 - June 2025*
Assistant Professor, Dept. of ECE

KTH Royal Institute of Technology, Sweden *Oct 2022 - Nov 2024*
Postdoctoral Researcher, School of EECS

EDUCATION

IDLab, Ghent University - imec *Oct 2017 - June 2022*
Doctorate in Computer Science Engineering
Thesis: Optimization Algorithms for Virtual Network and Media Services

Indian Institute of Technology Delhi *July 2015 - May 2017*
Master in Technology **GPA:** 9.588/10
Optoelectronics and Optical Communication
Thesis: Optical Frequency Shifters based on Stimulated Brillouin Scattering

National Institute of Technology Srinagar *2011 - June 2015*
Bachelor in Electronics and Communication Engineering **GPA:** 8.33/10

JOURNAL PUBLICATIONS

1. S. Egger, F. Dürr, B. Varga, MD Andrade, **GP Sharma**, J. Sachs, J. Harmatos, J. Gross, "Wireless-Aware TSN Engineering: Implications for 5G and Upcoming 6G Networks," , 2025. 📄
2. **G.P. Sharma** et al., "End-to-end Scheduling for Wired-wireless Mixed Networks," *Journal of Network and Systems Management*, 2024. 📄
3. **G.P. Sharma** et al., "Towards Deterministic Communications in 6G Networks: State of the Art, Open Challenges and the Way Forward," *IEEE Access*, 2023. 📄
4. **G.P. Sharma**, W. Tavernier, D. Colle, and M. Pickavet, "Routing and Scheduling for 1+1 Protected DetNet flows," *Computer Networks*, 2022. 📄
5. **G.P. Sharma**, W. Tavernier, D. Colle, and M. Pickavet, "Scheduling for Media Function Virtualization," *Future Internet*, vol. 13, no. 7, 2021. 📄
6. **G.P. Sharma**, D. Colle, W. Tavernier, and M. Pickavet, "On Decomposition and Deployment of Virtualized Media Services," *IEEE Transactions on Broadcasting*, vol. 67, no. 3, pp. 761–775, 2021. 📄
7. **G.P. Sharma**, W. Tavernier, D. Colle, and M. Pickavet, "VNF-AAPC: Accelerator-aware VNF Placement and Chaining," *Computer Networks*, vol. 177, 2020. 📄
8. **G.P. Sharma**, S. Preußler and T. Schneider, "Precise Optical Frequency Shifting Using Stimulated Brillouin Scattering in Optical Fibers," *IEEE Photonics Technology Letters*, vol. 29, no. 17, pp. 1467-1470, 1 Sept.1, 2017. 📄

CONFERENCE PUBLICATIONS

1. N. Roy, M.H. Dhullipalla, **G.P. Sharma**, D. Dimarogonas and J. Gross, “Quality of Control based Resource Dimensioning for Collaborative Edge Robotics,” Accepted at IEEE CCNC Workshops 2025 - ROBOCOM. 📄
2. S. Mostafavi, M. Tillner, **G.P. Sharma**, and J. Gross, “EDAF: An End-to-End Delay Analytics Framework for 5G-and-Beyond Networks,” Accepted at IEEE INFOCOM Workshops 2024 - CNERT. 📄
3. S. Mostafavi, V. Moothedath, S. Rönngren, N. Roy, **G.P. Sharma**, S. Seo, M.O. Muñoz and J. Gross, “ExPECA: An Experimental Platform for Trustworthy Edge Computing,” *ACM TEC*, 2023. 📄
4. S. Mostafavi, **G.P. Sharma**, and J. Gross, “Data-Driven Latency Probability Prediction for Wireless Networks: Focusing on Tail Probabilities,” *IEEE Globecom 2023*, Kuala Lumpur, Malaysia, 2023. 📄
5. J. Miserez, **G.P. Sharma** and W. Tavernier, “Routing protocols exploiting queue information for deterministic networks,” *International Conference on the Design of Reliable computer networks (DRCN)*, Vilanova, Spain, 2023. 📄
6. **G.P. Sharma**, D. Colle, W. Tavernier, and M. Pickavet, “Improving resource utilization with Virtual Media Function decomposition,” *International Conference on Multimedia Computing, Networking and Applications (MCNA)*, Valencia, Spain (virtual), 2020, pp. 31–37. 📄
7. **G.P. Sharma**, W. Tavernier, D. Colle, and M. Pickavet, “Hardware-accelerator aware VNF-chain recovery,” *International Conference on the Design of Reliable computer networks (DRCN)*, Milan, Italy (virtual), 2020. 📄
8. **G.P. Sharma**, D. Colle, W. Tavernier, and M. Pickavet, “VNF-AAP: Accelerator-aware Virtual Network Function Placement,” *IEEE Conference on Network Function Virtualization and Software Defined Networks (NFV-SDN)*, Dallas, USA, 2019. 📄
9. **G.P. Sharma**, W. Tavernier, D. Colle, and M. Pickavet, “Dynamic hardware-acceleration of VNFs in NFV environments,” *International Conference on Software Defined Systems (SDS)*, Rome, Italy, 2019, pp. 254–259. 📄
10. **G.P. Sharma**, W. Tavernier, D. Colle, and M. Pickavet, “Dynamic accelerator provisioning for SSH tunnels in NFV environments,” *IEEE Conference On Network Softwarization (Netsoft)*, Paris, France, 2019, pp. 242–244. 📄

PROJECTS

- Contributing to the Horizon 2020 EU-funded project **DETERMINISTIC6G** (2023-). Among other tasks, the key responsibility is to investigate data-driven methods to accurately characterize RAN latency to support time-critical applications.
- Worked on the problem of end-to-end scheduling for mixed wired-wireless TSN in the context of the FWO project **VERI-END** (2019-2022). The **optimization problem** for end-to-end packet scheduling was modeled as an ILP in the CPLEX environment and a greedy-based heuristic was proposed.
- Contributed to various tasks in the Horizon 2020 EU-funded project **NGPaaS** (2017-2019) led by NOKIA-BL Paris. On behalf of NGPaaS, gave a talk at India EU Stakeholders’ workshop on 5G Technology Landscape regarding the NGPaaS platform (Feb. 2019).
- Developed a scheme for dynamic provisioning of FPGA-based accelerator resources for VNFs in NFV environments.
- Worked on “**Optical Frequency Shifters based on Stimulated Brillouin Scattering**” (Aug 2016 - May 2017) as a part of my M.Tech thesis at IIT Delhi and TU Braunschweig.

ACHIEVEMENTS

- Recipient of the student travel grant to present a paper at the **IEEE NFV-SDN 2019** conference in Dallas, USA.
- Recipient of the **DAAD’s IIT Master Sandwich Scholarship 2016**
- Technical Manager of “**EMBESYS**”, an event organized under technical festival at NIT Srinagar in 2014
- Selected in the 18th **National Science Congress 2010** (National level), Chennai to present a project on rural water purification system

SKILLS AND COMPETENCIES

Programming languages	C, Python, MATLAB, Verilog
Tools	Git/Github, CPLEX, LabView

LANGUAGES

Hindi, English, German (A1)

PROFESSIONAL SERVICES

Journals Reviews

Comuter Networks, Computer Communications, IEEE Communication Letters, IEEE Access and

Conferences Reviews

IEEE ICCCN, IEEE DRCN, IEEE Globecom, IEEE Netsoft

INTERESTS

Badminton, Yoga, Cricket, Mindfulness

ONLINE COURSES AND MOOCS

- **Algorithms Specialization** offered by Coursera and Stanford University (April 2022)
- **Essentials of IP Media Transport for Broadcasters** offered by the SMPTE (February 2020)
- **Understanding SMPTE ST 2110** offered by the SMPTE (February 2020)
- **“Embedded Systems – Shape the World”** (May 2014) offered by edX and the University of Texas Austin (Achieved 100%)
- **Computer Networks** (April 2014) by Coursera and the University of Washington.

REFERENCES

Available on request.